Graph Theory

Question difficulty is indicated by 🌶 (lower difficulty), 🌶🌶 (greater difficulty), and 🌶🌶🌶 (greatest difficulty). A topic label is also given on the question.

Revision

Problem 1

2015 Kazakhstan National Olympiad 🌶

A full binary tree consists of $n+1$ levels, where from each vertex two edges go down and one edge comes from above (the top vertex at level 1 has no incoming edge, and vertices at the last $(n+1)$-th level have no outgoing edges).
In how many ways can the edges of this tree be coloured in 2^n given colours (each edge is coloured in one colour) so that for each colour, all edges of that colour form a path from some vertex to a vertex on the last level?

Solution 1

The tree has $n+1$ levels, meaning there are 2^n vertices at the last level. Each of these vertices has exactly one incoming edge from the level above, resulting in exactly 2^n edges entering the final level. Since each colour must form a single path that ends at a vertex on the last level, a path moving downwards cannot branch, meaning a single colour cannot end at more than one leaf. Since there are exactly 2^n available colours and 2^n edges entering the last level, every colour must be used exactly once at the bottom level. Thus, the 2^n edges entering the last level must be coloured using a permutation of the 2^n given colours. The number of ways to colour these bottom edges is $(2^n)!$.

Now, we move upwards from level n to the root. Each vertex at level n has 2 outgoing edges going down to the last level. These two edges already have distinct colours assigned from our first step. The single incoming edge to this vertex from level $n-1$ must continue the path of one of its colours. Therefore, it has exactly 2 choices: it must match the colour of either its left child edge or its right child edge.As we continue moving up to the root, every vertex at any level k (where $2 \le k \le n$) has:

- Two outgoing edges whose colours are already determined by the choices made below them. Because the sets of colours in the left and right subtrees are completely disjoint, these two outgoing colours are guaranteed to be distinct.
- One incoming edge from the level above, which must choose to match one of these 2 distinct outgoing colours.

Thus, every vertex except the root requires exactly a 2-way choice for its incoming edge. The total number of vertices that have an incoming edge from above (excluding the root and the bottom leaves) is the sum of vertices from level 2 to level n:

$$\text{Number of choices} = 2^1 + 2^2 + 2^3 + \cdots + 2^{n-1} = 2^n - 2$$

Since each of these vertices offers independent choices for its incoming edge, the number of ways to colour all the upper edges is 2^{2^n-2}. Thus the total number of ways is $(2^n)! \cdot 2^{2^n-2}$

Problem 2

2019 Canadian Mathematical Olympiad Problem 5

A 2-player game is played on $n \geq 3$ points, where no 3 points are collinear. Each move consists of selecting 2 of the points and drawing a new line segment connecting them. The first player to draw a line segment that creates an odd cycle loses. (An odd cycle must have all its vertices among the n points from the start, so the vertices of the cycle cannot be the intersections of the lines drawn.) Find all n such that the player to move first wins.

Solution 2

Let A and B denote the first and second player. We will prove that A wins with best play if and only if $n \equiv 2 \pmod 4$.

First of all, observe that the final state of the game is a complete bipartite graph $K_{a,b}$ with $a + b = n$. This means that A wins if only if a and b are both odd.

Define a balanced connected bipartite graph (on more than 1 vertex) to be one where both colour classes have equal cardinality. We also say a general bipartite graph is balanced if each nontrivial connected component (of more than 1 vertex) is balanced. Claim. If n is odd then B wins. Proof. If the final state is $K_{a,b}$, one of a and b must be even. Lemma. If n is even, then either player $S \in \{A, B\}$ can force the final state $K_{n/2,n/2}$. Proof. Let T be the other player.

The strategy for S is to ensure that the graph is always balanced after their turn (in particular, there are an even number of isolated vertices); certainly this is true at the outset. To see this is always possible, we take two cases on T's move.

If T joins an isolated vertex to a nontrivial connected component γ, then S can simply connect another isolated vertex to the opposite colour class in γ. Otherwise, T leaves the graph balanced. In this case, S may draw a new edge involving zero or two isolated vertices; this leaves the graph balanced.

Thus S can always ensure the graph is balanced on their turn, forcing the final state $K_{n/2,n/2}$. Corollary. If $n \equiv 2 \pmod 4$ then A wins. If $n \equiv 0 \pmod 4$ then B wins.

The desired conclusion follows.

Problem 3

IMO 2005

In a mathematical competition, 6 problems were posed to the participants. Every pair of problems was solved by more than $\frac{2}{5}$ of the contestants. Moreover, no contestant solved all six problems. Show that there are at least two contestants who solved exactly five problems each.

Solution 3

We can model this problem using graph theory, let each of the problems be a vertex, and an edge be a pairing of them solved.

6 problems so $C(6,2) = 15$ edges in total for a complete graph. Now assume each participant i solved k_i problems for $i = 1$ to n, then the number of ways they all solved every two problems is

$$\sum_{i=1}^{n} C(k_i, 2) > 15 \cdot \frac{2}{5} n = 6n$$

because more than $2/5$ of the n contestants got them.

Now to prove at least 2 solved exactly 5 problems, let us disprove where 0 or 1 solve 0 problems such that it has to be at least 2. If no one solved 5 problems then $k_i \leq 4$, so $C(k_i, 2) \leq 6$, so

$$\sum_{i=1}^{n} C(k, 2) \leq 6n$$

. This contradicts the $> 6n$ so someone must have solved 5 problems.

Assume one person solved 5 problems, so there is one $k_i = 5$ s.t. $C(k_i, 2) = C(5, 2) = 10$ while the rest have $C(k_j, 2) \leq 6$ where $j \neq i$ so

$$\sum_{i=1}^{n} C(k_i, 2) \leq 6(n-1) + 10 = 6n + 4.$$

As a result total solves between $6n+1, 6n+2, 6n+3, 6n+4$. Since $C(3,2) = 3$ and $C(4,2) = 6$, there must be at most one $k_i = 3$ person who only got 3 problems.

Now comes the hard part with this case, one person only solved 5 problems, potentially one person solved 3, the rest solved 4. We shall consider cases. If one person truly has $k_i = 3$ then

$$\sum_{i=1}^{n} C(k_i, 2) \leq 6(n-2) + 10 + 3 = 6n + 1,$$

which is less than the acceptable bound $6n + 3$ given we found total solves lie between $6n + 3$ and $6n + 4$. So there must be only 5 solver with the rest being 4 solvers. Disproving this is left as an exercise for the reader since it will require applications unrelated to graph theory.

As no one got 6 of them and 0 or 1 people can't get 5 of them, at least 2 must get 5 of them.

Cliques 🌶

Problem 4

Show that $K_{m,n}$ has at most a perfect square number of edges using Turán's theorem, where m, n are even and $m+n$ is constant.

Solution 4

$K_{m,n}$ has $m+n$ vertices so by Turan's theorem at most $\lfloor (m+n)^2/4 \rfloor$.

Let $m = 2k, n = 2l$ so $\lfloor (m+n)^2/4 \rfloor = \lfloor (4k^2 + 8kl + 4l^2)/4 \rfloor = \lfloor k^2 + 2kl + l^2 \rfloor \rfloor = \lfloor (k+l)^2 \rfloor$.

Matchings 🌶🌶

Problem 5

There are $2n$ people in a room, where each pair of people is classified as either friends or strangers. Two players play a game where they alternate turns picking a person who has not been picked before and is friends with the person picked on the previous turn.

The last player who can make a legal move wins. The first player may pick any person. Prove that the second player has a winning strategy if and only if the $2n$ people can be partitioned into n disjoint pairs such that the two people in each pair are friends.

Solution 5

This is a sketch of the solution:

Generalise this process as making matchings. For backwards direction, to prove if there is a perfect matching then player 2 wins we must assume there is a perfect matching. Then for every of the last player's pick you can pick a friend connected to that friend, so you always have best response so you win.

For forward direction, if second player wins then perfect matching. It is best to prove this by proving the contrapositive, if no perfect matching, then first player wins.

To prove this we apply Hall's condition, where if there is no perfect matching when the adjacent neighhbourhoods are not greater than the subset of friends to match. As a result the second player has a vertex or friend that cannot be matched, which means player 1 wins.

Problem 6

USA TSTST 2018 Problem 2 In the nation of Onewaynia, certain pairs of cities are connected by one-way roads. Every road connects exactly two cities (roads are allowed to cross each other, e.g., via bridges), and each pair of cities has at most one road between them. Moreover, every city has exactly two roads leaving it and exactly two roads entering it.

We wish to close half the roads of Onewaynia in such a way that every city has exactly one road leaving it and exactly one road entering it. Show that the number of ways to do so is a power of 2 greater than 1 (i.e. of the form 2^n for some integer $n \geq 1$).

Solution 6

In the language of graph theory, we have a simple digraph G which is 2-regular and we seek the number of sub-digraphs which are 1-regular. We now present two solution paths.

First solution, combinatorial We construct a simple undirected bipartite graph Γ as follows:

the vertex set consists of two copies of $V(G)$, say V_{out} and V_{in}; and for $v \in V_{\text{out}}$ and $w \in V_{\text{in}}$ we have an undirected edge $vw \in E(\Gamma)$ if and only if the directed edge $v \to w$ is in G.

Moreover, the desired sub-digraphs of H correspond exactly to perfect matchings of Γ.

However the graph Γ is 2-regular and hence consists of several disjoint (simple) cycles of even length. If there are n such cycles, the number of perfect matchings is 2^n, as desired.